

# Vasanth P

## Data Scientist

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### PROFESSIONAL SUMMARY

Aspiring **Data Scientist** skilled in **Machine Learning**, **Deep Learning**, and **AI-driven solutions**. Strong analytical mindset with a passion for building data-driven applications and interdisciplinary problem-solving.

### SKILLS

**Statistical Programming:** — Python, **Data Mining** — clustering, association rules, anomaly detection, **SQL / Querying Relational Databases** — MySQL, **Visualization:** — PowerBi, Excel & Tableau, **Statistical Methods** — Logistic Regression, A/B Testing, Hypothesis Testing, Feature Engineering., **Predictive Modeling** — Regression, Classification, Time Series Forecasting, NLP, **Cloud:** — AWS, **Advanced Tools** — Scikit-learn, XGBoost, Pandas

### EDUCATION

#### Master of Computer Application

Manipal University of Jaipur

12/2025 – Present | Salem, Tamil Nadu

#### IIT-M Advanced Programming Professional & Master Data Science

IIT-M GUVI HCL

10/2024 – 10/2025 | Salem

#### Bachelor's of Arts in Hisotry

Bharathidasan University

06/2016 – 05/2019 | Trichy, Tamil Nadu

### CERTIFICATES

Master Data Science Program 📄 , Python zero to hero 📄 , My SQL 📄 , Data Analysis and Processing with Python using Pandas 📄 , Probability and Statistics for Data Science 📄 , Introduction to Machine Learning 📄 , Deep Learning Fundamentals 📄

### PROJECTS

#### EcoPulse – PM2.5 Nowcasting for Smarter Cities 📄

Live PM2.5 Nowcast and Forecast with Streamlit

Objective: Developed an AI-driven analytics system to nowcast and forecast PM2.5 air quality across Indian cities using real-time data.

Key Contributions:

- Engineered an end-to-end ML pipeline integrating **OpenAQ** and **Open-Meteo APIs** for 94% real-time data coverage.
- Designed and optimized regression models with feature engineering on weather and temporal variables, improving forecast reliability by **18%**.
- Evaluated model performance using **MAE (10.2  $\mu\text{g}/\text{m}^3$ )** and  **$R^2$  (0.87)** benchmarks.
- Deployed an interactive Streamlit dashboard providing city-wise air quality insights and anomaly alerts.
- Tech Stack:** Python, Machine Learning, API Integration, Streamlit, Data Engineering, Time Series Analysis

#### Brain Tumor Classification using Deep Learning 📄

Objective: Created an AI-powered diagnostic system for accurate MRI-based brain tumor classification.

Key Contributions:

- Built and trained multiple CNN architectures including **ResNet50** and **MobileNet**, achieving **93% accuracy** and reducing misclassification by **18%**.
- Implemented data augmentation and preprocessing for over **7,000 MRI scans**, improving model generalization.
- Fine-tuned hyperparameters in **TensorFlow/Keras** for better convergence and inference speed.
- Deployed a Streamlit-based prediction interface for real-time tumor classification.
- Tech Stack:** Python, TensorFlow, Keras, Deep Learning, Transfer Learning, Image Preprocessing, Model Evaluation, Streamlit

#### Shopper Spectrum

Customer Segmentation and Product Recommendations in E-Commerce

Key Contributions:

- Architected an AI-based engine to analyze customer purchase behavior and improve personalization.
- Applied **RFM analysis** and **K-Means clustering** to segment customers, driving actionable insights.
- Built **collaborative filtering** model, improving product relevance by **22%**.
- Deployed via Streamlit web app, enhancing decision-making efficiency.
- Tech Stack:** Machine Learning, Clustering (K-Means), Collaborative Filtering, RFM Analysis, Feature Engineering, Streamlit, Data Visualization, EDA, Scikit-learn, Python